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SUPERSEDING
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MILITARY SPECIFICATION

CONVERTERS, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for devices which convert from a nominal 115/200 Volt, 400 Hertz, 3-phase, alternating current (ac) (see 6.4.1) aircraft electrical system to nominal 28 Volt unregulated direct current (dc) (see 6.4.4) power.

1.2 Classification. Converters shall be of the following classes, as specified (see 6.2):

Class A: Converters which are self-cooled and conform to the ambient temperature-altitude requirements of MS33543, Curve II, except that no operation shall be required below -55°C.

Class B: Converters which are self-cooled and conform to the ambient temperature-altitude requirements of MS33543, Curve I, except that the maximum ambient temperature shall not exceed +85°C and no operation shall be required below -55°C.

Class C: Converters which are blast-cooled and conform to the ambient temperature-altitude and inlet airflow requirements specified in the applicable detail specification, and converters which are self-cooled and conform to the temperature-altitude requirements of MS33543, Curve I, except that no operation shall be required below -55°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-W-5088	Wiring, Aerospace Vehicle
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-T-7928	Terminals, Lug: Splices, Conductor: Crimp-Style, Copper, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-A-21180	Aluminum-Alloy Castings, High Strength
MIL-C-39006/22	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR79
MIL-C-39006/25	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR81
MIL-R-39015	Resistors, Variable, Wire-Wound (lead screw actuated) Established Reliability, General Specification for
MIL-I-46058	Insulating Compound, Electrical (for coating printed circuit assemblies)

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MIL-E-81910 Electrical Power Generating and Control Equipment, Aircraft, General Specification for

STANDARDS

Federal

FED-STD-595 Color

Military

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

MIL-STD-198 Capacitors, Selection and Use of

MIL-STD-454 Standard General Requirements for Electronic Equipment

MIL-STD-481 Configuration Control-Engineering Changes (Short Form), Deviations and Waivers

MIL-STD-785 Reliability Program for Systems and Equipment Development and Production

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

MIL-STD-831 Test Reports, Preparation of

MIL-STD-970 Standards and Specifications, Order of Preference for the Selection of

MIL-STD-1562 List of Standard Microcircuits

MIL-STD-1629 Procedures for Performing a Failure Mode, Effects and Criticality Analysis

MIL-STD-2000 Standard Requirements for Soldered Electrical and Electronic Assemblies

MIL-STD-2073-1 DOD Materiel Procedures for Development and Application of Packaging Requirements

MIL-STD-2073-2 Packing Requirement Codes

MIL-STD-2175 Castings, Classification and Inspection of

MS14151 Washer, Flat, Round, Corrosion Resisting Steel (for Electrical Applications)

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MS33543	Criteria-Temperature and Altitude Range Self-Cooled Electric Equipment
MS35338	Washer, Lock-Spring Helical, Regular (Medium) Series, (m/mm)
MS35650	Nut, Plain - Hexagon, Machine Screw, UNF-2B

(See Supplement 1 of this specification for list of associated Military Detail Specifications.)

MILITARY HANDBOOKS

MIL-HDBK-217	Reliability of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL AIR SYSTEMS COMMAND DOCUMENTS

AS-4613	Application and Derating Requirements for Electrical Components

(Copies are available from the Naval Air Systems Command, (Attn: AIR-5112), Jefferson Plaza 2, Washington, DC 20361-5110.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications (see 6.4.3)), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual converter requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between requirements of this specification and the applicable detail specification, the latter shall govern. (If a specific requirement specified herein is not required for a converter, it shall be so indicated on the associated detail specification; for example, "Shock - Not applicable.").

3.2 Classification of requirements. The requirements for converters are classified herein as follows:

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Requirement	Paragraph
Qualification	3.3
Materials	3.4
Selection of parts and materials	3.5
Design and construction	3.6
Performance characteristics	3.7
Reliability	3.8
Markings	3.9
Workmanship	3.10

3.3 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.5 and 6.3). Each unit delivered shall fully meet all the requirements of this specification and the applicable detail specification, and shall be capable of successfully completing the tests required by this specification for quality conformance inspection and for inspections listed for any one of the qualification samples. Any changes in the design of the converter shall be handled as specified in MIL-STD-481, with copies of the documentation provided to both the qualifying activity and, if applicable, the procuring activity.

3.4 Materials.

3.4.1 Metals. Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

3.4.1.1 Dissimilar metals. Unless otherwise suitably protected, dissimilar metals such as brass, copper, or steel shall not be used in intimate contact with aluminum or its alloys. When protection is used it shall be of such a type that a low impedance path is offered to radio-frequency currents. Dissimilar metals are defined as two metal specimens that generate an electric current in a conductive solution when in contact or otherwise electrically connected.

3.4.1.2 Magnesium alloy parts. Neither magnesium nor magnesium alloy parts shall be used.

3.4.1.3 Finishes and protective coatings.

3.4.1.3.1 Aluminum alloy parts. Aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the areas under screws, nuts, etc. used for assembly or mounting purposes.

3.4.1.3.2 Steel parts. Steel parts shall be plated, if necessary, so that the converter meets all of the specified electrical and environmental requirements.

3.4.1.3.3 Color. Equipment shall be finished in a color conforming to FED-STD-595, color number 37038 (black). The finish shall be unaffected by environmental testing required by the applicable detail specification.

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3.4.1.4 Castings. Castings shall be in accordance with MIL-STD-2175 and shall be Class 3, Grade C or better. Aluminum alloy castings shall be in accordance with MIL-A-21180. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnating, welding, or other methods acceptable to the acquiring activity. Inspection of repaired castings shall be governed by quality control techniques subject to the approval of the acquiring activity.

3.4.2 Nonmetallic materials. Nonmetallic materials used shall be nontoxic, fungus inert, flame resistant and shall not support combustion.

3.5 Selection of parts and materials. Specifications and standards for all materials and parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the design and construction of converters complying with the requirements of this specification shall be selected in accordance with MIL-STD-970.

3.5.1 Standard parts. Military standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the applicable drawings by their military part numbers. Any standard part, of the same part number, shall be readily interchangeable as applied within the converter.

3.5.2 Non-standard parts. The use of non-standard parts is permitted in the event that there is no suitable standard part available on the date of invitation for bids, however, approval prior to use shall be obtained from the qualifying activity. If a non-standard part is approved by the qualifying activity, subsequent approval is not necessary. Request for approval for the use of non-standard parts shall be submitted in writing and shall contain the following information:

a. Designation of equipment and assembly wherein part is used.

b. Reference designation of part (circuit symbol, part number, as applicable).

c. Name and complete detailed technical description of part.

d. Part manufacturer's name and address.

e. Part manufacturer's drawing and part number.

f. Reason for not using standard part including, where applicable, a comparison of characteristics of the non-standard part which most closely possesses the required characteristics.

g. Other information required herein for specific types of parts.

3.5.3 Commercial (hardware) parts. Commercial utility hardware items: screws, bolts, nuts, and cotter pins may be used without approval provided they possess suitable properties and are replaceable by MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and on the contractor's drawings.

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3.5.4 Semiconductor devices. Military standard semiconductor devices shall be chosen, applied and reported in accordance with Requirement 30 of MIL-STD-454 and shall be selected from those meeting MIL-S-19500 and listed in QPL-19500. The reliability level of devices selected shall be in the following order of preference beginning with the most preferred: JANTXV, JANTX, JAN. Requests for approval to use non-standard semiconductors shall be accompanied by substantiating reliability data or proposed testing to assure adequate device reliability for the application.

3.5.5 Microelectronic devices. Microelectronic devices shall be chosen, applied and reported in accordance with Requirement 64 of MIL-STD-454 and shall be listed in MIL-STD-1562. Requests for approval to use devices which do not comply with this requirement shall be submitted as non-standard parts and shall be accompanied by substantiating reliability data or proposed testing to assure adequate device reliability for the application.

3.5.6 Capacitors. Capacitors shall be selected and applied in accordance with MIL-STD-198. The use of wet slug tantalum capacitors, except tantalum cased units in accordance with MIL-C-39006/22 and MIL-C-39006/25, requires the approval of the qualifying activity and, if applicable, the procuring activity. Silver cased tantalum capacitors shall not be used.

3.5.7 Electron tubes. Electron tubes of either the vacuum or gas filled type shall not be used.

3.5.8 Potentiometers and adjustable resistors. Only sealed units conforming to MIL-R-39015 shall be used unless approved as a non-standard part.

3.6 Design and construction. The converter shall be a single package and shall be a completely static solid state unit having no moving parts, except as noted in 3.6.2.7. Dynamic protective devices (fuses, circuit breakers and thermally actuated switches) shall not be used unless specified in the applicable detail specification. Any design changes to the converter shall be handled as specified in 3.3.

3.6.1 Electrical.

3.6.1.1 Component derating. All converter electric/electronic components shall be derated as shown in Table I based upon the most severe converter environmental and operational conditions. Stress levels in excess of the above shall be submitted to the qualifying activity for written approval along with the rationale for the use of that stress level in the application.

3.6.1.2 Electric connections.

3.6.1.2.1 External electric connections. The equipment shall be so designed that external connections and wiring can be made in accordance with the wiring installation, ground return, and connectors requirements of MIL-W-5088. The equipment structure shall not be used as a current path except for radio noise shielding.

3.6.1.2.1.1 Terminal block design. Where terminals are used for the external connection to converters, they shall be of the stud-type, and shall be designed

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so that the current is conducted by means of surface-to-surface contact and not through the stud threads. All stud threads shall be steel, corrosion resistant, C-34 Rockwell minimum hardness, and shall be of sufficient length to accommodate two MIL-T-7928 lugs. Nuts and washers used on terminals shall conform to MS35650 for nuts, MS35338 for lockwashers, and MS14151 for flat washers. There shall be no dielectric material in the compression build-up. The terminal block shall be so designed that it can be removed and replaced on the converter without the necessity of rebrazing or soldering. Barriers affording a positive separation of leads and terminals shall be provided on the terminal block. Durable, reusable, nonconductive terminal covers shall be provided. Terminal designations shall be durably, legibly, and prominently marked on the terminal block.

3.6.1.2.2 Internal electric connections. Internal connections shall be made in accordance with MIL-STD-454 Requirements 1, 10, 17, 19, 20, and 69. General purpose soldering shall be performed in accordance with MIL-STD-2000 Task A. For applications which include active surface mounted microelectronic devices, soldering shall be performed in accordance with MIL-STD-2000 Task F.

3.6.2 Mechanical.

3.6.2.1 Maintainability. The converter shall be designed to facilitate assembly, disassembly, location of trouble sources, and maintenance without the aid of special tools. All functional parts shall be readily identifiable, accessible, and removable for replacement. When printed circuit boards are utilized, only clear transparent conformal coating conforming to MIL-I-46058, Types UR or XY, shall be used in order to permit identification, location, removal, and replacement of functional parts. Functional parts are defined as individual component parts such as resistors, capacitors, semiconductor devices, or replaceable functional assemblies such as active or passive filters, voltage dividers, dual transistors, or diodes.

3.6.2.2 Failure mode and effects analysis (FMEA). A failure mode and effects analysis shall be performed in accordance with the requirements of MIL-STD-1629 Task 101.

3.6.2.3 Shock mounts. Shock mounts or vibration isolators shall not be used.

3.6.2.4 Interchangeability. Provisions for interchangeability are required to the maximum extent possible. Design tolerances shall permit parts, subassemblies, and assemblies to be used in the converter without regard to the source of supply or manufacturer. Parts, subassemblies, and assemblies having the full range of dimensions and characteristics permitted by the governing specification shall be usable as replacement items without selection and without departure from the specified performance requirements of the converter. When permission is granted to use a non-standard part or material because the existing standard part or material is not available, the converter shall be so designed that the non-standard part or material and the standard part or material are interchangeable. When the specification for the part or material contains substitutability or supersession information, the design shall permit the substitute or superseding parts or materials to be used interchangeably.

3.6.2.5 Screw threads. Screw threads shall conform to the requirements of MIL-S-7742.

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3.6.2.6 Safety wiring and staking. Accidental loosening of screws and screw parts, and other connections, shall be prevented by safety wiring (.032 inch minimum OD) where practicable, staking, self locking nuts, or other approved methods. Washers and cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.6.2.7 Cooling. No conduction cooling to the converter environment shall be allowed. Converters self-cooled by internal fan shall utilize a motor having ball bearings of quality level ABEC-3 or higher, and the fan shall be rated for operation at 100°C ambient or higher. Converters which are cooled by an external blast air supply shall meet the requirements of this specification and the applicable detail specification when supplied with cooling air flow and temperature defined by the applicable detail specification. Pressure drop through the unit shall meet the requirements of the applicable detail specification.

3.6.2.8 Ventilation openings. All ventilation or other openings, except for the air inlet for blast-cooled converters, shall be designed to prevent passage of foreign objects of one-fourth inch diameter or larger.

3.6.2.9 Operating position. The converter shall operate in any position.

3.6.2.10 Size and weight. The dimensions and weight shall be as specified in the applicable detail specification. The weight and dimensions shown on the applicable detail specification shall include the weight and dimensions of all auxiliary apparatus necessary to make the converter conform to this specification and the applicable detail specification. Where only maximum and minimum dimensions are shown, the part need not have the shape shown. But, the part, including all protrusions, shall be contained within the outline shown.

3.7 Performance characteristics. The converter shall deliver power in accordance with the requirements specified herein and the applicable detail specification.

3.7.1 Electrical.

3.7.1.1 Input power. The converter shall meet the requirements of this specification and the applicable detail specification when supplied with power as defined herein. Nominal input voltage is three-phase, wye-connected, 115 Volts alternating voltage (see 6.4.2) at 400 Hertz, input phase rotation in the order A, B, C, phase displacement of 120° ± 4° and waveform and input voltage unbalance having full performance characteristics defined in 3.7.1.1.1c., d., and e. below.

3.7.1.1.1 Steady state input voltage.

3.7.1.1.1.1 Full performance. The converter shall meet full performance output requirements (see 3.7.1.2.1.1) with any combination of the following input characteristics:

a. Steady state input voltage within the range defined in the applicable detail specification.

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b. Steady state input frequency within the range defined in the applicable detail specification.

c. Input waveform total distortion of 5 percent of the fundamental or less, with individual frequency components (including modulation sidebands) within the limits of figure 1, curve A.

d. Input voltage unbalance of 3 Volts or less between phase voltages.

e. Input voltage amplitude modulation of 2.5 Volts RMS.

3.7.1.1.1.2 Limited performance. The converter shall meet limited performance output requirements (see 3.7.1.2.1.2) when each of the following characteristics is separately substituted for the corresponding characteristic listed for full performance listed in 3.7.1.1.1.1:

a. Steady state input voltages between 108 Volts RMS and the minimum specified in the applicable detail specification.

b. Steady state input voltages between the maximum input specified in the applicable detail specification and 118 Volts RMS, when the maximum input specified is less than 118 Volts RMS.

c. Input waveform total distortion of 8 percent of the fundamental or less, with individual frequency components (including modulation sidebands) within the limits of figure 1, curve B.

3.7.1.1.2 Transient input voltage.

3.7.1.1.2.1 Transient surge susceptibility.

3.7.1.1.2.1.1 Full performance. The converter output shall remain within the limits of figure 2A when subjected to transient input voltages within the limits of figure 2B.

3.7.1.1.2.1.2 Limited performance. The converter shall not be damaged by power interrupts or abnormal input surge undervoltages from 0 to 102 Volts RMS. The converter shall not be damaged by abnormal input surge overvoltages from 124 to 180 Volts RMS. (see 4.7.2.12.3.1.2)

3.7.1.1.2.2 Transient spike susceptibility. The converter shall meet full performance output requirements specified in 3.7.1.2.1.1 when subjected to input spike voltages as shown in figure 6 (see 4.7.2.12.3.2).

3.7.1.1.3 Transient input frequency. The converter shall withstand without damage and shall recover full performance after application of the following frequency transients (see 4.7.2.12.4):

480 Hz for 7 seconds
320 Hz for 7 seconds

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The converter output voltage shall remain between 22.5 Volts and the maximum voltage allowed by the applicable detail specification during application of the following frequency transients:

450 Hz for 2 seconds
440 Hz for 15 seconds
360 Hz for 15 seconds
350 Hz for 2 seconds

3.7.1.1.4 Input power factor. The converter input power factor shall be no less than 0.9 lagging when operated from 50 percent to 100 percent rated load (see 4.7.2.5).

3.7.1.1.5 Input current balance. The current input to each phase of the converter shall remain within 5 percent of the average of the three-phase currents from 25 percent rated load to 100 percent rated load (see 4.7.2.6).

3.7.1.1.6 Total harmonic distortion. When operating from minimum to full rated load, the converter shall not drive the total harmonic distortion of the input voltage above 5 percent when powered by a source with total harmonic distortion of 3.5 percent or less into a purely resistive load.

3.7.1.1.6.1 Input current waveform. The converter shall not cause single harmonic line currents that are greater than 3 percent of the converter's full rated load fundamental current from the second to the thirty-second harmonic. Currents with frequencies from the thirty-second to the fiftieth harmonic shall not exceed 100/n percent of the converter's full rated load fundamental current, where n is the harmonic multiple number.

3.7.1.2 Output power.

3.7.1.2.1 Steady state output voltage. The indicated output characteristics shall apply from minimum to full rated output load current.

3.7.1.2.1.1 Full performance. Steady state output voltage characteristics shall be as follows under the conditions listed in 3.7.1.1.1.1:

a. Steady state output voltage shall be within the limits specified in the applicable detail specification.

b. Peak to mean ripple voltage for either polarity shall not exceed 1.5 Volts.

c. Output distortion factor shall not exceed .035 (output distortion factor is the ratio of the RMS value of the ripple to the average dc voltage).

d. Output ripple component frequencies shall not exceed the limits of figure 3, curve A.

3.7.1.2.1.2 Limited performance. Steady state output voltage characteristics shall be as follows under the conditions listed in 3.7.1.1.1.2:

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a. With the input specified in 3.7.1.1.1.2a., the output voltage shall be as specified in the applicable detail specification.

b. With the input specified in 3.7.1.1.1.2b., the output voltage shall be as specified in the applicable detail specification.

c. With input waveform specified in 3.7.1.1.1.2c., peak to mean output ripple voltage for either polarity shall not exceed 2.0 Volts, nor shall output ripple component frequencies exceed the limits of figure 3, curve B.

3.7.1.2.2 Shock load recovery. When the following loads are applied and removed, output voltage transients shall remain within the limits of figure 2A: 100 percent rated load current, 200 percent rated load current, and short circuit (see 4.7.2.7).

3.7.1.2.3 Input power application. The output voltage when the converter is initially energized shall be within steady state limits within one second after application of the input voltage. During the one second power-up period, output voltage transients shall be within the limits of figure 2A. For determining compliance of voltage transients to figure 2A, the onset of the transient shall be considered to be the time when the output voltage first reaches a value equal to minimum steady state output voltage specified in the applicable detail specification (see 4.7.2.8).

3.7.1.2.4 Overload and short circuit capacity. Over an input voltage range of 112.5 to 117.5 Volts RMS and input frequency range of 393 to 407 Hertz, the converter shall deliver overload and short circuit current, without damage, for the duration specified in figure 4. With loads of 500 percent rated current or less, the converter shall maintain the minimum output voltage specified in figure 5 (see 4.7.2.9).

3.7.1.2.5 Parallel operation. The converter shall operate with its output connected in parallel with one to three other converters of the same manufacturer's part number when each unit is being supplied from the same power source having full performance characteristics specified in 3.7.1.1.1.1. The load carried by each converter shall not differ from another by more than 10 percent of the unit's rating with system loads from 10 percent to rated system load. Rated system load is the sum of the ratings of all individual units. Required output power characteristics shall be the same as for a single converter (see 4.7.2.10).

3.7.1.3 Efficiency. When supplied from an input power source having full performance characteristics as specified in 3.7.1.1.1.1, converter efficiency shall be 80 percent minimum from 25 percent to 50 percent rated load, and 85 percent minimum from 51 percent to full rated load (see 4.7.2.3).

3.7.1.4 Electromagnetic compatibility. The converter shall meet the electromagnetic interference requirements of MIL-E-81910 (see 4.7.2.11).

3.7.1.5 Acoustical noise. When limits are specified by the applicable detail specification, and when tested as specified in 4.7.2.13, the maximum sound level measured at any point shall not exceed the specified limits (see 4.7.2.13).

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3.7.1.6 Pressure drop (blast-cooled converters only). Pressure drop through blast-cooled converters shall not exceed the values specified in the applicable detail specification under the conditions defined therein (see 4.7.2.14).

3.7.2 Environmental.

3.7.2.1 Temperature-altitude. The converter shall meet all performance requirements specified herein throughout the temperature-altitude range specified in 1.2 for the class specified in the applicable detail specification.

3.7.2.2 Acceleration. When this test is required by the applicable detail specification, the converter shall meet the requirements specified herein when subjected to the acceleration test of MIL-E-81910 (see 4.7.2.16).

3.7.2.3 Humidity. The converter shall meet the requirements specified herein when subjected to the humidity test of MIL-E-81910 (see 4.7.2.17).

3.7.2.4 Shock. The converter shall meet the requirements specified herein when subjected to the mechanical shock test of MIL-E-81910. The converter shall be operated during the basic design test. Any output voltage transients resulting from the shock shall remain within the limits of figure 2A (see 4.7.2.18).

3.7.2.5 Vibration. The converter shall meet the requirements specified herein when subjected to the vibration test of MIL-E-81910 (see 4.7.2.19).

3.7.2.6 Fungus. The converter shall meet the requirements specified herein when subjected to the fungus test of MIL-E-81910 (see 4.7.2.20).

3.7.2.7 Salt fog. The converter shall meet the requirements specified herein when subjected to the salt fog test of MIL-E-81910 (see 4.7.2.21).

3.7.2.8 Dust. The converter shall meet the requirements specified herein when subjected to the dust test of MIL-E-81910 (see 4.7.2.22).

3.7.2.9 Explosive atmosphere. The converter shall not cause ignition when tested as specified in 4.7.2.23.

3.8 Reliability. Reliability shall be considered in each phase of the converter design process. Techniques used to determine and evaluate the detailed design shall include a stress analysis with associated predictions and allocations in conformance with MIL-HDBK-217; sneak circuit and worst case analysis on all converter control and protective circuits in accordance with MIL-STD-785 Tasks 205 and 206; and parts derating in accordance with the requirements of AS-4613 for class B equipment. Converters shall have a minimum observed Mean Flight Hours Between Failure (MFHBF) of 10,000 hours when operated under specified conditions.

3.8.1 Reliability demonstration. The reliability of the converter shall be demonstrated by a test, analyze, and fix (TAAF) test program (see 4.7.2.24).

3.9 Markings.

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3.9.1 Terminals. Markings of external electrical connections shall be as specified on the applicable detail specification.

3.9.2 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.9.3 Nameplate. A nameplate conforming to MIL-P-15024, Type A, B, C, G, or H, with the following markings, in addition to the requirements of MIL-STD-130, shall be securely attached to the converter. The nameplate shall contain the following information:

Converter _____ Ampere
Input: _____ Volts AC _____ Amperes/Phase AC
Output: _____ Volts DC _____ Amperes DC

Military Part No.
National Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Date of manufacture
Contract or Order No.
Manufacturer's name or trademark
Inspector's stamp
US

3.9.4 Airflow. The direction of airflow in blast-cooled or fan-cooled units shall be plainly and permanently marked with an arrow and the word "airflow". When the direction of airflow is marked on the cover, provision shall be made to assure that the cover cannot be installed to indicate a false airflow direction.

3.10 Workmanship. The converter shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All converters shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements,

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however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Qualification retention inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 Qualification inspection.

4.4.1 Qualification tests. Qualification tests shall be conducted at a facility satisfactory to the activity responsible for qualification (see 6.3). Qualification tests shall consist of the examination, inspections, and tests specified in Table II, in the order shown or as otherwise approved by the qualifying activity.

4.4.2 Qualification samples. The qualification test sample shall consist of at least four (4) units of each part number for which qualification is sought. The sample units shall be produced, to the maximum extent practicable, with equipments and procedures normally used in production, and shall be identified as specified (see 3.9). Any additional information required by the letter of authorization shall be provided (see 6.3). The samples shall have successfully completed the quality conformance inspections shown in Table III.

4.4.3 Data submission.

4.4.3.1 Drawings. Together with the test samples, the manufacturer shall submit two (2) reproducible copies of the outline and detail assembly drawings, schematics, and parts lists. Upon completion of qualification testing and prior to qualification approval, the manufacturer shall again provide two (2) reproducible copies of the outline and detail assembly drawings, schematics, and parts lists. Any changes in the reproducible copies from those submitted with the qualification samples shall be indicated in detail.

4.4.3.2 Certification. Together with the test samples, the manufacturer shall submit a materials inspection report consisting of a certification, supported by verifying data, such that the materials and finishes specified in 3.4 and used in fabricating the converters are in accordance with the applicable referenced specification or requirements prior to such fabrication. Also, the manufacturer shall submit documentation certifying that the converter meets the explosive atmosphere requirements specified in 3.7.2.9.

4.4.3.3 Component stress level. Together with the test samples, the manufacturer shall submit a report listing all converter electric/electronic components, their maximum rated stress, their applied stress within the

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converter, and the ratio of applied to rated stress. An analysis of electric/electronic components shall be performed in accordance with MIL-HDBK-217 to demonstrate conformance to paragraph 3.8 of this specification. Data shall be based upon the most severe converter environmental and operational conditions specified herein and the applicable detail specification.

4.4.4 Disposition of qualification samples and test report. Upon completion of the qualification tests, the test samples shall be identified as required and forwarded to the qualifying activity (see 3.3). The test samples shall be accompanied by three (3) copies of the qualification test report, prepared in accordance with MIL-STD-831.

4.4.5 Retention of qualification. To retain qualification, the manufacturer shall comply with the requirements of 4.5. In the event that no production occurred during a twelve (12) month period, a report shall be submitted to the qualifying activity certifying that the company still has the capabilities and facilities necessary to produce the item. If during two (2) consecutive twelve (12) month periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product sample for testing in accordance with the qualification inspection requirements.

4.5 Qualification retention inspections. Each converter of the same part number delivered as a qualified item under this specification shall be numbered sequentially in essentially the order that it is submitted for Government acceptance, regardless of the contract under which it is delivered or by whom it is purchased. To meet the delivery requirements for the first 50 or less converters, the contract delivery requirement plus one additional converter must successfully complete the required quality conformance inspections. The contract delivery of required converters are to be shipped as directed by the applicable contract(s). The additional converter (retention sample), selected at random by the Government Inspector, is to be forwarded to the Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304 (Reference: 101). To meet the delivery requirements for each succeeding 150 converters, 151 converters must successfully complete the required quality conformance inspection, 150 of which are to be shipped as directed by the applicable contract(s). The additional converter (retention sample) shall be randomly selected and forwarded as above. The span of serial numbers and the contract number of the production units from which each retention sample was selected shall be recorded, and this information included with the retention sample. Each retention sample shall successfully complete the quality conformance inspection shown in Table III, and the test data and test date shall also be included with the converter (retention sample). The samples shall be shipped no later than the final unit in the production group from which they are selected. The units forwarded to the qualifying activity will be inspected and tested for conformance to this specification and the applicable detail specification. Tests will be conducted for each converter (retention sample) according to any column from Table II at the discretion of the qualifying activity. All retention samples become Government property when accepted by the Government Inspector for shipment to the qualifying activity. The cost for each converter (retention sample) shall be borne by the manufacturer. These requirements shall apply to all converters procured in accordance with this specification whether procured directly by the Government or by private industry procuring for end item use by the Government.

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4.5.1 Qualification retention samples. For each quotation and based on all production contracts obtained, it will be the manufacturer's responsibility to inform the procuring activity of the exact number of qualification retention samples required for forwarding to the qualifying activity. The number of qualification retention samples required should be taken into account when the contract bid is prepared. If the proper number of retention samples is not itemized in the contract, then the manufacturer will, without additional cost to the Government, forward to the qualifying activity whatever number of retention samples is determined by the cognizant Government Inspector to be required by paragraph 4.5 of this specification.

4.5.2 Rejection and retest of qualification and qualification retention samples. Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification retention tests, may be reworked or have parts replaced to correct defects. The manufacturer shall bear the cost of any repair or redesign work necessitated by a failure to meet the specified requirements. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. After the second attempt to meet one of the requirements of this specification or the applicable detail specification, the manufacturer shall bear the cost of any additional retesting for that requirement. Where qualification tests are conducted under the auspices of the manufacturer, the qualifying activity will be advised upon failure of a qualification sample and the action taken by the manufacturer with regard to the failure. If the manufacturer does not correct and return to the qualifying activity converters rejected during qualification or qualification retention testing within ten (10) days after the manufacturer receives them or does not submit a program acceptable to the qualifying activity for the correction of deficiencies noted within ten (10) days, the manufacturer will be notified of the proposed cancellation of the qualification program or proposed removal of the item from the Qualified Products List, as appropriate.

4.6 Quality conformance inspections. Quality conformance inspections shall consist of inspection of the product for delivery. Each converter offered for delivery shall be subjected to the inspection specified in Table III. Data from these inspections shall be recorded and be made available for examination by authorized Government representatives.

4.6.1 Failure. Any converter failing to comply with specified requirements during the quality conformance inspections shall not be delivered on the contract or purchase order. The manufacturer may take corrective action on the failed converters as appropriate and resubmit for inspection. Reworked units shall be suitably marked to identify them as repaired items. In the event of failure during reinspection, further rework and resubmission shall not be allowed.

4.7 Methods of inspection.

4.7.1 Test conditions. Unless otherwise specified, tests shall be made under the following conditions.

4.7.1.1 Ambient temperature. Tests shall be performed at an ambient temperature of 25 ± 15°C.

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4.7.1.2 Ambient altitude. Tests shall be run at approximately sea level altitude.

4.7.1.3 Tolerance and accuracy. The tolerance of test conditions, other than room ambient, and the accuracy of test apparatus shall be as specified in the test facilities requirements of MIL-E-81910.

4.7.1.4 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above, the converter, and through averaging the results. For units without fans, air temperature measurements shall be taken 1 ± 1/4 inch from the surface of the converter. For units with an internal fan, air temperature measurements taken 1 ± 1/4 inch from the fan inlet shall be considered the ambient temperature.

4.7.1.5 Airflow. For all maximum temperature-altitude tests below 20,000 feet conducted on self-cooled converters without integral cooling fans, the airflow velocity over the converter within the test chamber shall be adjusted to approximate natural convection conditions and shall not exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance. Unless otherwise specified, inlet airflow during blast-cooled converter tests shall be maintained at nominal conditions as defined in the applicable detail specification.

4.7.1.6 Stabilization. Stabilization of test conditions based on constant input voltage and ambient temperature is defined to have occurred when the change over a ten (10) minute period is measured to be less than:

a. Output voltage: 0.02 Volts
b. Converter critical component temperature: 0.5°C

4.7.1.7 Voltage measurement. The input and output voltage shall be measured at the terminals of the unit. For three wire input converters with no neutral connection to the converter, the neutral measurement connection is made at the power source.

4.7.1.8 Position. The units shall be mounted base down.

4.7.1.9 Loading. Prior to testing, the converter shall be operated at full rated load (see 6.4.5) until stabilized. The test shall be performed with the converter operating at full rated load.

4.7.1.10 Input voltage. Nominal input voltage shall be applied.

4.7.2 Method of test.

4.7.2.1 Visual and mechanical inspection. The converter shall be inspected to determine compliance with the requirements of this specification and the applicable detail specification with respect to materials, design and construction, parts, workmanship, dimensions, weight, and markings, for which there are no applicable tests.

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4.7.2.2 Voltage regulation. The output voltage shall be measured under the conditions specified in 4.7.2.2.1 through 4.7.2.2.5. For each load, input voltage, position, temperature-altitude (and for blast-cooled converters inlet airflow condition), the output voltage shall remain within the limits specified in 3.7.1.2.1.1. During the course of the tests no components of the converter shall exceed their maximum rated temperatures. Tests may be conducted at any intermediate input voltage, load, or temperature-altitude condition at the option of the activity responsible for qualification.

4.7.2.2.1 Voltage regulation at room ambient temperature-altitude. Immediately upon energizing the converter proper operation shall be verified. The following tests, a. through e. below, shall be conducted after operating the converter at nominal input (see 3.7.1.1) and full load, until the converter is fully stabilized (see 4.7.1.6).

a. With nominal input voltage specified in 3.7.1.1 the load shall be varied from minimum load to full load in steps of 25 percent rating. When a load between minimum and 25 percent is specified on the applicable detail specification, the converter shall also be tested at this load. For blast-cooled converters only, during this portion of the test at full load, the input airflow conditions shall be varied from nominal conditions throughout the entire range defined by the applicable detail specification to establish the airflow conditions which result in maximum and then minimum internal component temperatures of the converter when stabilized at full load.

b. The test specified in 4.7.2.2.1a shall be repeated at the maximum input voltage specified by the applicable detail specification, and at 118 Volts RMS.

c. The test specified in 4.7.2.2.1a shall be repeated at the minimum input voltage specified by the applicable detail specification, and at 108 Volts RMS.

d. The tests specified in 4.7.2.2.1a, b, and c shall be repeated at the maximum input frequency specified by the applicable detail specification.

e. The tests specified in 4.7.2.2.1a, b, and c shall be repeated at the minimum input frequency specified by the applicable detail specification.

4.7.2.2.2 Voltage regulation at maximum temperature-altitude conditions. The converter shall be tested under each of the temperature-altitude conditions listed below. Prior to being energized, the converter shall be soaked at the initial maximum temperature, sea level condition for at least one-half hour. The test specified in 4.7.2.2.1 shall be conducted. Inlet conditions for blast-cooled converters shall be those determined to result in the highest internal component temperatures of the converter during the test of 4.7.2.2.1a.

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Class A Converters		Class B Converters	
Ambient Temperature°C	Altitude (Feet)	Ambient Temperature°C	Altitude (Feet)
+71	Sea Level	+85	Sea Level
+25	15,000	+85	17,500
0	30,000	+50	35,000
-10	50,000	+20	50,000
		+20	65,000

Class C Converters (blast-cooled)		Class C Converters (self-cooled)	
Ambient Temperature°C	Altitude (Feet)	Ambient Temperature°C	Altitude (Feet)
Maximum specified on the applicable detail specification	Sea Level	+120	Sea Level
	25,000	+70	25,000
	50,000	+20	50,000
	65,000	+20	65,000

4.7.2.2.3 Voltage regulation at minimum temperature-altitude conditions. The test specified in 4.7.2.2.2 shall be repeated substituting the following temperature-altitude conditions and increasing the soak time at the initial -55°C sea level condition to two hours. Inlet airflow conditions for blast-cooled converters shall be those determined to result in the lowest internal component temperatures of the converter during the test of 4.7.2.2.1a.

Class A, B, and C Converters

Ambient Temperature°C	Altitude (Feet)
-55	Sea Level
-55	10,000
-55	25,000
-55	65,000

4.7.2.2.4 Operating position. The converter shall be operated at nominal input and full load in the normal base down position until output voltage and critical component temperatures stabilize (see 4.7.1.6). Output voltage and critical component temperatures shall be measured as the converter is rotated by increments of 90 degrees from the normal base down position, 360 degrees about

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each major axis except the vertical axis. Each position shall be held until output voltage and critical component temperatures stabilize.

4.7.2.2.5 Worst condition. The converter shall be operated at the worst indicated conditions determined from the tests specified in 4.7.2.2.1 through 4.7.2.2.4. The conditions from each of these tests resulting in the highest output voltage shall be simultaneously applied. Prior to the test, the converter shall be soaked at the applicable ambient for a minimum of two hours prior to energizing the converter. The converter shall be operated until output voltage and critical component temperatures stabilize. The above test shall be repeated for minimum output voltage and maximum critical component temperature conditions.

4.7.2.3 Efficiency. In conjunction with the test of 4.7.2.2.1, the input and output power of the unit shall be measured. The efficiency shall be calculated under these conditions and shall not be less than specified in 3.7.1.3 under stabilized test conditions as defined in 4.7.1.6.

4.7.2.4 Output ripple. The output peak to mean ripple voltage shall be measured for both the positive and negative polarities during all tests of 4.7.2.2, and shall not exceed the limits specified in 3.7.1.2.1.1. The output ripple voltage component frequencies and distortion factor shall be measured during the test of 4.7.2.2.1 and shall not exceed the limit specified in 3.7.1.2.1.1.

4.7.2.5 Input power factor. In conjunction with the test of 4.7.2.2.1, the input power factor at half load or greater shall be determined for each condition and shall be as specified in 3.7.1.1.4.

4.7.2.6 Input current balance. In conjunction with the test of 4.7.2.2.1, the input current balance at 25 percent load or greater shall be determined for each condition and shall be as specified in 3.7.1.1.5.

4.7.2.7 Shock load recovery. The converter shall be tested to verify compliance with 3.7.1.2.2 at maximum, minimum, and room ambient temperatures for the applicable class converter at sea level altitude. For 100 percent load application and removal, tests shall be conducted at nominal input and minimum and maximum input voltages specified by the applicable detail specification. For 200 percent load and short circuit application and removal, tests shall be conducted at nominal input voltage, 112.5 Volts RMS input, and 117.5 Volts RMS input.

4.7.2.8 Input power application. The converter shall be tested to demonstrate compliance with input power application requirements of 3.7.1.2.3 at minimum, half, and full rated load at nominal input and maximum and minimum input voltages specified by the applicable detail specification. Tests shall be conducted at maximum, minimum, and room ambient temperature for the applicable class converter at sea level altitude.

4.7.2.9 Overload and short circuit capacity. The converter shall be tested to demonstrate compliance with the overload and short circuit requirements of 3.7.1.2.4 at maximum, minimum, and room ambient temperatures at sea level altitude for the applicable class converter. Tests shall be conducted at nominal input voltage and at the extremes of input voltage and frequency in 3.7.1.2.4.

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Tests shall be conducted at the test points indicated on figure 4. Should the 1200 percent rated current test point not be obtainable because of converter output impedance limitations, a test at the highest current obtainable shall be conducted for the corresponding duration indicated on figure 4.

4.7.2.10 Parallel operation. Two, three, and four identical converters shall have their outputs connected in parallel and the load varied from 10 percent of total rated capacity in steps of 10 percent. Converters shall be tested at nominal input and maximum and minimum input voltages and maximum and minimum input frequencies specified in the applicable detail specification when supplied by the same power source. The parallel converters, both individually and as a whole, shall meet the requirements of 3.7.1.2.5.

4.7.2.11 Electromagnetic compatibility. The converter shall be tested to demonstrate compliance with 3.7.1.4 at full and half load with nominal input and maximum and minimum input voltage specified by the applicable detail specification.

4.7.2.12 Input power susceptibility.

4.7.2.12.1 Input voltage unbalance. The test of 4.7.2.2.1a shall be conducted substituting each of the following six input voltage combinations for nominal input voltage.

Input Voltage (Volts RMS)

Combination Number	Phase A	Phase B	Phase C
1	113	116	116
2	116	113	116
3	116	116	113
4	113	113	116
5	113	116	113
6	116	113	113

During the test, output voltage, ripple voltage, distortion factor, and output ripple component frequencies shall meet the requirements of 3.7.1.2.1.1.

4.7.2.12.2 Input waveform distortion. The test of 4.7.2.2.1a shall be conducted at half and full rated load under the following conditions:

a. Using an ac power source which normally provides 3.0 percent total distortion or less while supplying the converter at full load, sufficient input inductance shall be added equally in each line (except the neutral line, if utilized) between the power source and the converter to increase input waveform distortion to the maximum allowed by 3.7.1.1.1.1c. During this test, output voltage, ripple voltage, distortion factor and output ripple component frequencies shall meet the requirements of 3.7.1.2.1.1. Input inductance shall then be further increased to obtain the maximum distortion allowed by 3.7.1.1.1.2c. During this test, output voltage, ripple voltage, and output ripple component frequencies shall meet the requirements of 3.7.1.2.1.1a and 3.7.1.2.1.2c.

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b. Using a variable waveform ac power source, input waveform distortion shall be adjusted to the maximum allowed by 3.7.1.1.1.1c while maintaining a crest factor of 1.31 to 1.40. This test shall be repeated while maintaining a crest factor of 1.42 to 1.51. During these tests, output voltage, ripple voltage, distortion factor and output ripple component frequencies shall meet the requirements of 3.7.1.2.1.1. Distortion shall then be adjusted to the maximum allowed by 3.7.1.1.1.2c while maintaining a crest factor of 1.26 to 1.38. This test shall be repeated while maintaining a crest factor of 1.44 to 1.56. During these tests, output voltage, ripple voltage, and ripple component frequencies shall meet the requirements of 3.7.1.2.1.1a and 3.7.1.2.1.2c.

4.7.2.12.3 Transient input voltage.

4.7.2.12.3.1 Transient surge susceptibility.

4.7.2.12.3.1.1 Full performance. With the converter operating at minimum load and full load with nominal input, the following positive step voltages shall be applied to the converter input terminals four times for each condition:

180 Volts RMS for 0.010 seconds
165 Volts RMS for 0.029 seconds
145 Volts RMS for 0.054 seconds
130 Volts RMS for 0.073 seconds
95 Volts RMS for 0.048 seconds
80 Volts RMS for 0.010 seconds

The converter output shall remain within the limits of figure 2A. Following the test, the converter shall pass the test of 4.7.2.15.

4.7.2.12.3.1.2 Limited performance. With the converter operating at minimum load and full load with nominal input, the following positive step voltages shall be applied to the converter input terminals four times for each condition:

180 Volts RMS for 0.1 seconds
162 Volts RMS for 0.5 seconds
124 Volts RMS for 25.0 seconds
102 Volts RMS for 25.0 seconds
64 Volts RMS for 0.1 seconds
5 Volts RMS for 0.05 seconds
0 Volts RMS for 0.05 seconds

The converter shall be observed for evidence of flashover or insulation breakdown, and shall meet the requirements of 3.7.1.1.2.1. Following the test, the converter shall pass the test of 4.7.2.15.

4.7.2.12.3.2 Transient spike susceptibility. The converter shall be operated at minimum, half, and full load at the maximum input voltage specified in the applicable detail specification. Positive and negative input spike voltages synchronized to the positive peak, negative peak, and the zero crossing of each input phase voltage shall be applied to the converter input terminals for 5 minutes for each condition.Spikes shall have a waveform as shown in figure 6 when generated into an open circuit, and shall have a source impedance of 50 ± 5 Ohms. The spike waveform shall be recorded at the converter terminals during the test.

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The output of the converter shall be monitored to verify compliance with 3.7.1.1.2.2.

4.7.2.12.4 Transient input frequency. The converter shall be tested to demonstrate compliance with 3.7.1.1.3. Frequency transients shall be applied exponentially with a 0.030 second time constant. Following the test, the converter shall pass the test of 4.7.2.15.

4.7.2.13 Acoustical noise. The acoustical noise of the converter shall be measured in planes one foot from each of the six sides. In each plane, measurements shall be made opposite the center of the converter surface to determine the sound level. The maximum sound level at all points shall be measured using the B weighted scale of an integrating-type sound level meter. Microphone orientation shall be in accordance with the recommendations of the manufacturer of the sound level meter to obtain the required frequency response. The calibrating devices or standards used shall be directly traceable to the National Institute of Science and Technology. Measurements shall be made under anechoic conditions with the converter supported in a manner to minimize acoustic transmission and reflection by the support structure. Compliance to 3.7.1.5 shall be determined.

4.7.2.14 Pressure drop (blast-cooled converters only). The requirements of 3.7.1.6 shall be verified by test.

4.7.2.15 Performance verification. During and following tests where satisfactory converter operation is to be verified, the test of 4.7.2.2.1 shall be conducted except that both positive and negative peak to mean ripple voltages shall also be measured at each condition. Blast-cooled converters shall be tested with nominal inlet airflow conditions. Output voltage and ripple voltage shall meet the requirements of 3.7.1.2.1.1.

4.7.2.16 Acceleration. The converter shall be tested in accordance with the acceleration test of MIL-E-81910. The converter shall be operated at nominal input and one-fourth load during Procedure II, and the output voltage and ripple voltage shall also be monitored during the test to determine compliance with 3.7.1.2.1.1. At the conclusion of the test, the converter shall be examined for evidence of damage, and shall be tested as specified in 4.7.2.15.

4.7.2.17 Humidity. The converter shall be tested in accordance with the humidity test of MIL-E-81910. Periods of operation during the test shall be at nominal input and 10 percent load. During this operation, output voltage and ripple voltage shall be monitored and comply with 3.7.1.2.1.1. Following the test, the converter shall pass the test of 4.7.2.15.

4.7.2.18 Shock. The converter shall be tested in accordance with the mechanical shock test of MIL-E-81910, except that the converter shall be operating during the basic design test at full load and nominal input. The output voltage shall be monitored during the test to determine compliance with 3.7.2.4. At the conclusion of the test, the converter shall be inspected for evidence of damage and the unit shall pass the test specified in 4.7.2.15. The converter shall not be operated during the crash safety test.

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4.7.2.19 Vibration. The converter shall be tested in accordance with the vibration test of MIL-E-81910. During the test the converter shall be operated with nominal input voltage and full rated load. Blast-cooled converters may be operated with no inlet blast air at the maximum load condition specified in the applicable detail specification for the no inlet blast air condition. The output voltage and ripple voltage shall be monitored for compliance with 3.7.1.2.1.1. Following the test, the converter shall pass the test of 4.7.2.15.

4.7.2.20 Fungus. The converter shall be tested in accordance with the fungus test of MIL-E-81910. At the conclusion of the test, the converter shall show no evidence of fungus growth, and shall pass the test of 4.7.2.15.

4.7.2.21 Salt fog. The converter shall be tested in accordance with the salt fog test of MIL-E-81910. Specified drying shall be defined as allowing the unit to stand base down at room ambient conditions for 24 hours. The converter shall be inspected for evidence of corrosion and related damage. The unit shall pass the test of 4.7.2.15. This test shall be repeated after 48 hours additional exposure at room ambient conditions.

4.7.2.22 Dust. The converter shall be tested in accordance with the dust test of MIL-E-81910. The converter shall be operated at no load with nominal input voltage, when required. At the conclusion of the test, the converter shall pass the test of 4.7.2.15. For units equipped with a cooling fan, proper operation (speed and noise) shall be verified.

4.7.2.23 Explosive atmosphere. The converter shall be tested in accordance with MIL-STD-810, Method 511.3, Procedure I.

4.7.2.24 Reliability demonstration. The reliability demonstration test shall be structured as a test, analyze, and fix (TAAF) program wherein any failure (i.e. any operation outside of the specified limits) which occurs is analyzed, and corrective action implemented by the manufacturer to prevent recurrence of the failure. A minimum of 3,000 hours total converter operating time on 2 samples shall be accumulated with one or less failures under the test profile conditions on figure 7 to successfully complete the test. In the event that more than one failure occurs, additional converter operating time shall be accumulated until the total converter operating hours divided by the number of failures equals 3,000 or higher. When a corrective action has resolved a failure to the satisfaction of the qualifying activity, all but the first occurrence of the resolved failure shall be dropped from consideration. No single converter of the two test samples shall accumulate less than 25 percent of the total test time. During the test, output voltage and ripple shall be monitored for compliance with 3.7.1.2.1.1a and b. The test is conducted at full rated load except at 500 ± 50 hour intervals, when at such times the output load of each test sample shall be increased to 400 percent rated load for 7 seconds during the +71°C portion of the cycle. During these overloads, the converter shall meet the requirements of 3.7.1.2.4. Following the complete test, the converter shall pass the following tests: 4.7.2.2.1, 4.7.2.2.2 (at sea level only), 4.7.2.2.3 (at sea level only), 4.7.2.3, 4.7.2.4, 4.7.2.5, 4.7.2.6, 4.7.2.9.

4.7.2.25 Disassembly and inspection. The converter shall be disassembled and inspected for defects or deterioration which could prevent the converter from

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meeting the requirements of this specification. Such defects or deterioration shall be sufficient cause for rejection.

5. PACKAGING

5.1 Preservation and packaging. The converters shall be packaged in accordance with MIL-STD-2073-1, level A or C, as specified in the contract or order.

5.1.1 Level A. For level A packaging, preservation shall be in accordance with MIL-P-116, Method II, without preservation compounds, using metal reusable containers unless otherwise specified.

5.1.2 Level C. For level C packaging, the converter shall be preserved and packaged individually in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be in accordance with MIL-STD-2073-1, level A, B, or C, as specified in the contract or order.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-2073-2 and MIL-STD-129.

5.3.1 Reinspection dates. Reinspection date marking shall be as specified by the acquiring activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items described herein are intended for use on military aircraft to convert three-phase, 115 VRMS, 400 Hertz AC power to nominal 28 Volt unregulated DC power. Converters typically are used to provide DC power on aircraft with main electrical generating systems that provide AC power.

6.2 Acquisition requirements.

a. Title, number, and date of this specification.

b. Class.

c. Military part number.

d. Required level of packaging and packing.

e. Type of container.

f. Reinspection date marking, and any special marking.

g. Handbooks or manuals, when required.

h. That the manufacturer shall include the cost of units required for qualification retention (see 4.5) in the amount of his bid.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for the opening of bids, qualified for inclusion in the applicable Qualified Products List - whether or not such products have actually been so listed by that date. The attention of the manufacturers is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command (AIR-5363), Department of the Navy, Washington, D.C. 20361-5360; however, information pertaining to qualification of products may be obtained from the qualifying activity - the Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304.

6.3.1 Qualification retention samples. For each quotation, and based on all production contracts obtained, it will be the manufacturer's responsibility to inform the procuring activity of the exact number of qualification retention samples required for forwarding to the qualifying activity (see 4.5). The number of qualification retention samples required should be taken into account when the contract bid is prepared. If the proper number of retention samples is not itemized in the contract then the manufacturer will, without additional cost to the Government, forward to the qualifying activity whatever number of retention samples is determined by the cognizant Government Inspector to be required by paragraph 4.5 of this specification.

6.4 Definitions.

6.4.1 Alternating current. Alternating currents used in this specification shall be root mean square (RMS) values.

6.4.2 Alternating voltage. Alternating voltages used in this specification shall be considered line-to-neutral root mean square (RMS) values for both 3 wire and 4 wire input connections.

6.4.3 Detail specification. Detail specification used in this specification shall be either an MS Drawing or Military Specification Sheet, as applicable, detailing the requirements for a specific converter.

6.4.4 Direct voltage and current. Direct voltages and currents used in this specification shall be mean (average) values.

6.4.5 Full rated load (full load). Full rated load used in this specification shall be the maximum continuous converter output load current specified in the applicable detail specification (see 6.4.3) for a given temperature-altitude condition.

6.5 Installation. The aircraft manufacturer shall allow clearance of at least one inch at each surface of the converter (except the mounting surface) in addition to the maximum dimensions shown on the applicable detail specification for installation and ventilation.

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6.6 Subject term (key word) listing.

Electrical
Power Supply
Transformer-rectifier
Unregulated

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Table I. Component derating.

Part Type	Critical Stress	Maximum Allowable Application Stress
MICROCIRCUITS		
Digital	Output Current	.80
	Operating Frequency	.75
Linear	Bias Voltage	.70
	Input Signal Voltage	.70
	Output Current	.75
	Operating Frequency	.75
Voltage Regulator	Input Voltage (Rated Maximum)	.80
	Output Current (Rated Maximum)	.75
	Power Dissipation (Rated Maximum)	.60
TRANSISTORS		
General Purpose	Power	.50
	Current	.75
	Voltage	.75
Power	Power	.30
	Current	.75
	Voltage	.75
Switching	Power	.50
	Current	.75
	Voltage	.75
DIODES		
General Purpose	Power	.50
	PIV	.50
	Surge Current	.50
	FWD Current	.50
Rectifier	Power	.30
	PIV	.50
	Surge Current	.50
	FWD Current	.50

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Table I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
DIODES		
Switching	Power	.30
	PIV	.50
	Surge Current	.50
	FWD Current	.50
SCR	Power	.30
	PIV	.50
	Surge Current	.50
	FWD Current	.50
Varactor	Power	.50
	PIV	.75
	FWD Current	.75
Zener	Power	.50
	FWD Current	.50
	Zener Current	$.5[I_z$ max + I_z nom]
Reference	Power	.30
	FWD Current	.50
	Zener Current	$.5[I_z$ max + I_z nom]
CAPACITORS, FIXED		
Ceramic	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Mica, Dipped	Voltage	.60
	Current	
	AC/Ripple	.70
	Surge	.70
Glass	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70

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TABLE I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
CAPACITORS, FIXED		
Porcelain	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Paper	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Plastic	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Tantalum, Foil or Film	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Tantalum, Solid, Dry	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
	Protection	3 OHMS/VOLT
Aluminum Electrolytic	Voltage	.65
	Current	
	AC/Ripple	.70
	Surge	.70

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TABLE I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
CAPACITORS, ADJUSTABLE		
Ceramic	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Glass	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Air	Voltage	.30
RESISTORS, FIXED		
Carbon Composition	Power	.50
Insulated Film	Power	.50
Wirewound Precision		
1.0% Tol.	Power	.50
.1% Tol.	Power	.25
Wirewound Power	Power	.50
Thermistor	Power	.50
RESISTORS, ADJUSTABLE		
Wirewound	Rated Current	.70
Non Wirewound	Rated Current	.70

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TABLE I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
RESISTORS, ALL	Voltage	.80
MISCELLANEOUS		
Transformers and Chokes	Flux Saturation level	.75
	Insul. Breakdown Volt.	.50
	Current	.70

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TABLE II. Qualification inspection.

INSPECTION	CONVERTER SAMPLE				REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
	1	2	3	4		
Visual & mechanical inspection	X	X	X	X	3.4, 3.5, 3.6, 3.9, 3.10	4.7.2.1
Voltage regulation					3.7.1.1.1.1, 3.7.1.2.1.1	4.7.2.2
Room ambient temp-alt	X	X	X	X		4.7.2.2.1
Maximum temp-alt	X	X				4.7.2.2.2
Minimum temp-alt	X	X				4.7.2.2.3
Operating position	X	X				4.7.2.2.4
Worst condition	X	X				4.7.2.2.5
Efficiency	X	X	X	X	3.7.1.3	4.7.2.3
Output ripple	X	X	X	X	3.7.1.1.1.1, 3.7.1.2.1.1	4.7.2.4
Input power factor	X	X	X	X	3.7.1.1.4	4.7.2.5
Input current balance	X	X	X	X	3.7.1.1.5	4.7.2.6
Total harmonic distortion	X	X	X	X	3.7.1.1.6	---
Shock load recovery	X		X		3.7.1.2.2	4.7.2.7
Input power application	X		X		3.7.1.2.3	4.7.2.8
Overload & short circuit capacity		X	X		3.7.1.2.4	4.7.2.9
Parallel operation	X	X	X	X	3.7.1.2.5	4.7.2.10
Electromagnetic compatibility	X		X		3.7.1.4	4.7.2.11
Input power susceptibility		X			3.7.1.1.1.2, 3.7.1.1.2, 3.7.1.1.3, 3.7.1.2.1.2	4.7.2.12
Acoustical noise	X				3.7.1.5	4.7.2.13
Pressure drop	X				3.7.1.6	4.7.2.14
Acceleration	X				3.7.2.2	4.7.2.16
Humidity	X				3.7.2.3	4.7.2.17
Shock	X				3.7.2.4	4.7.2.18
Vibration		X		X	3.7.2.5	4.7.2.19
Fungus				X	3.7.2.6	4.7.2.20
Salt fog		X			3.7.2.7	4.7.2.21
Dust				X	3.7.2.8	4.7.2.22
Explosive atmosphere		X			3.7.2.9	4.7.2.23
Reliability demonstration			X	X	3.8	4.7.2.24
Disassembly & inspection			X	X	---	4.7.2.25

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TABLE III. Quality conformance inspection.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Visual & mechanical inspection	3.4, 3.5, 3.6, 3.9, 3.10	4.7.2.1
Overload & short circuit capacity (at room ambient conditions, nominal input voltage and frequency only. Converters shall be tested at 200% and 500% test points only.)	3.7.1.2.4	4.7.2.10
Voltage regulation at room ambient temperature-altitude	3.7.1.2.1.1	4.7.2.2.1 a, b, c only
Efficiency	3.7.1.3	4.7.2.3
Output ripple	3.7.1.1.1.1, 3.7.1.2.1.1	4.7.2.4
Input power factor	3.7.1.1.4	4.7.2.5
Input current balance	3.7.1.1.5	4.7.2.6
Total harmonic distortion	3.7.1.1.6	---

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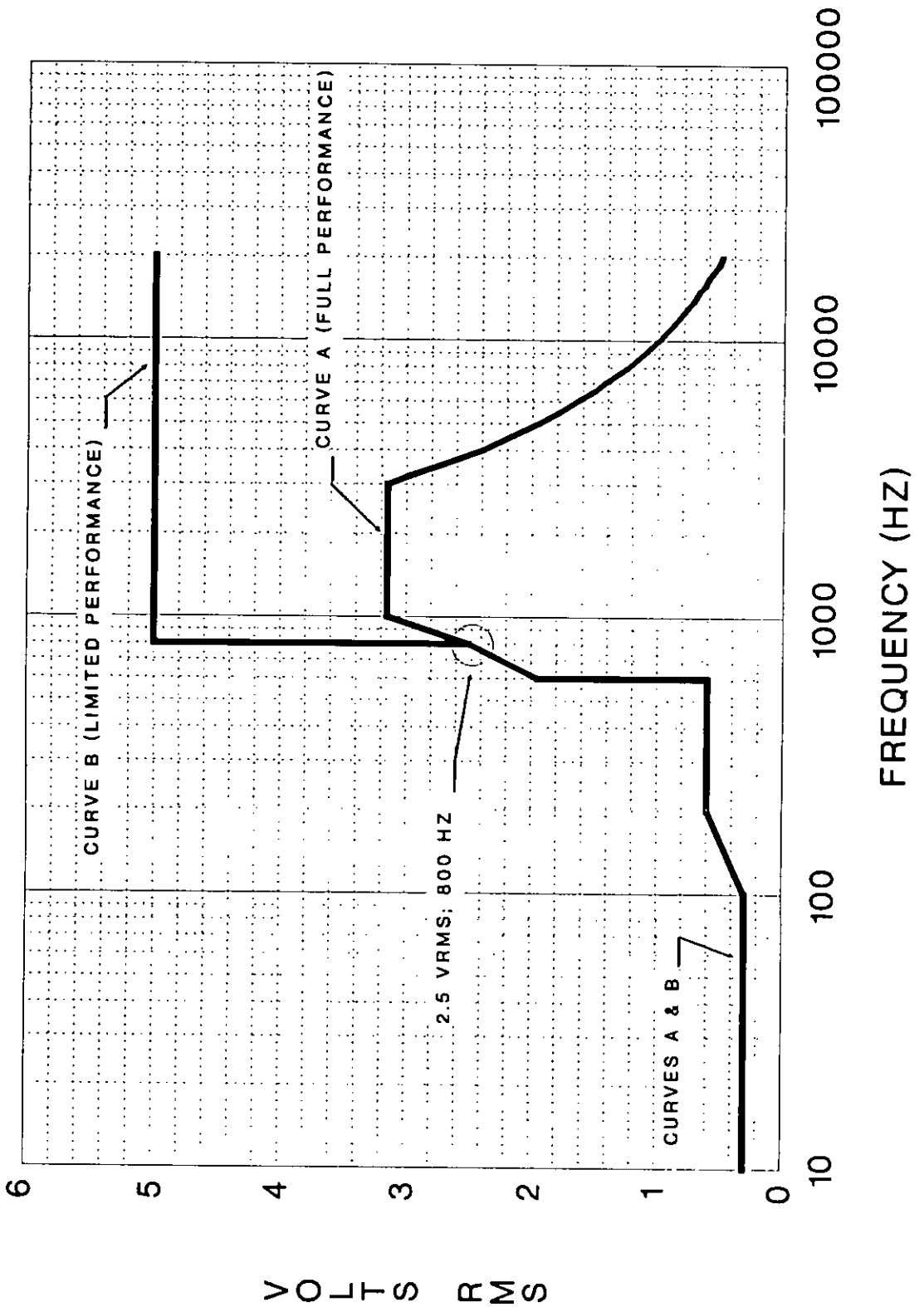


Figure 1. Input waveform distortion.

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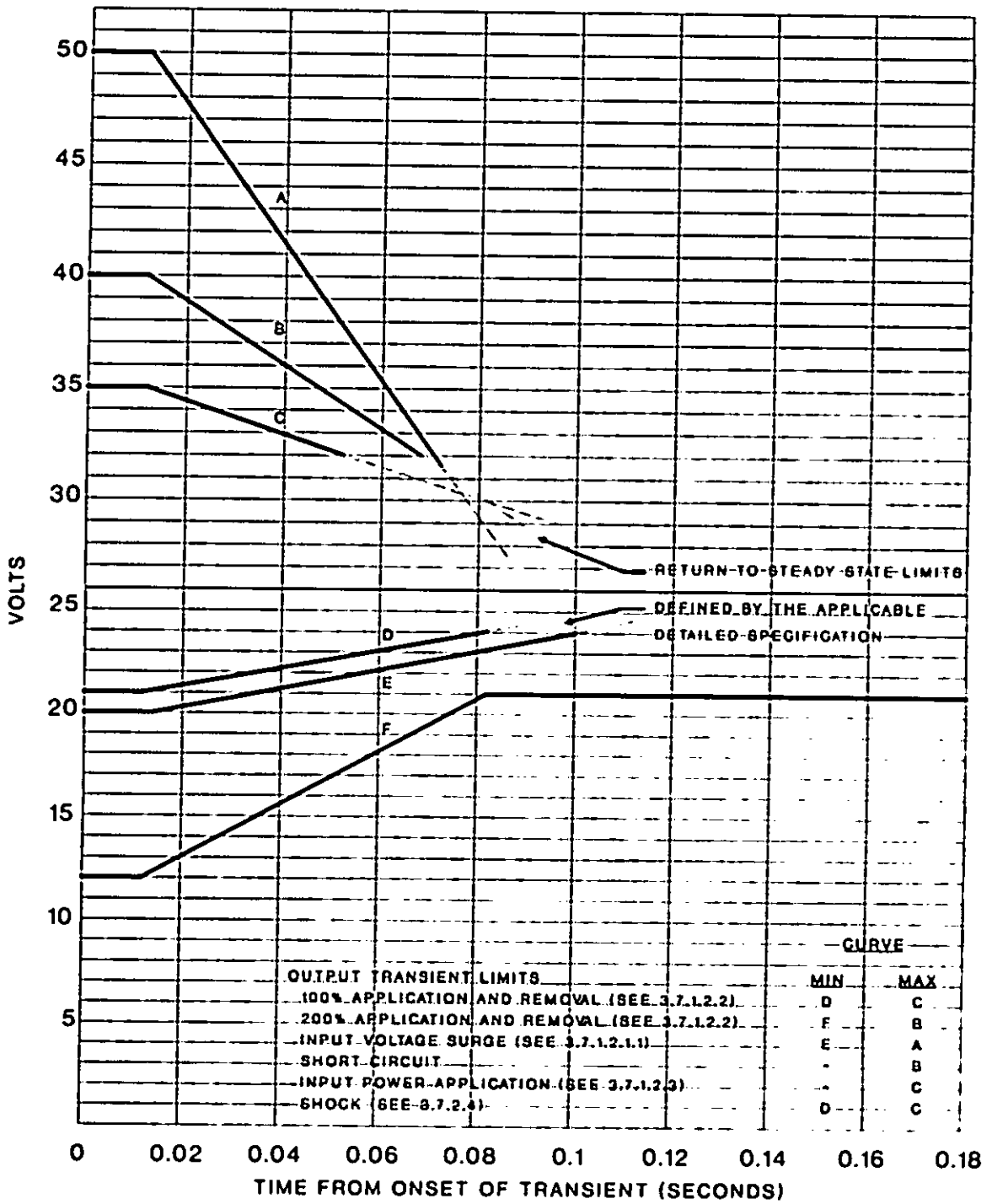


Figure 2A. Output voltage transient limits.

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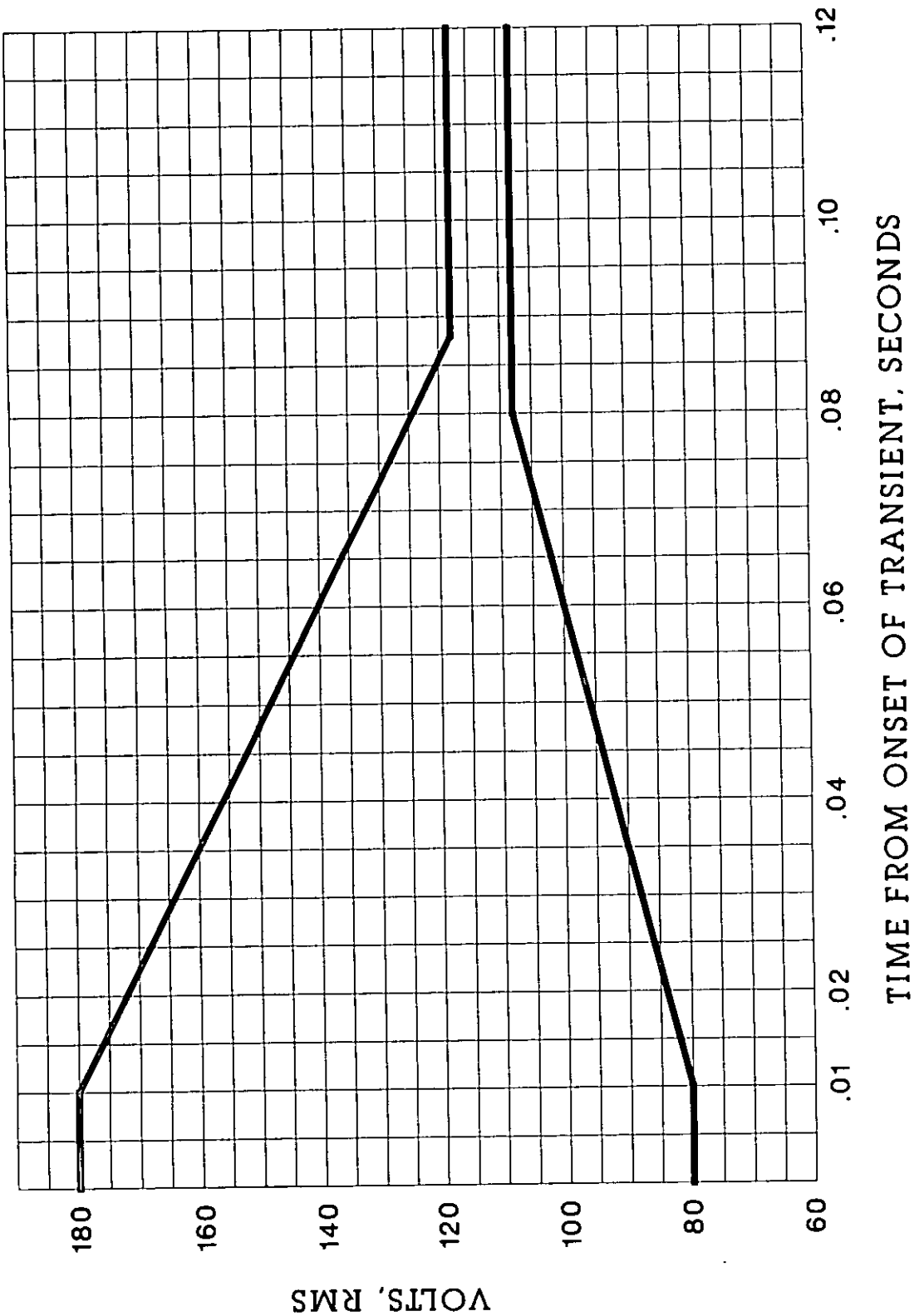


Figure 2B. Input voltage transient limits.

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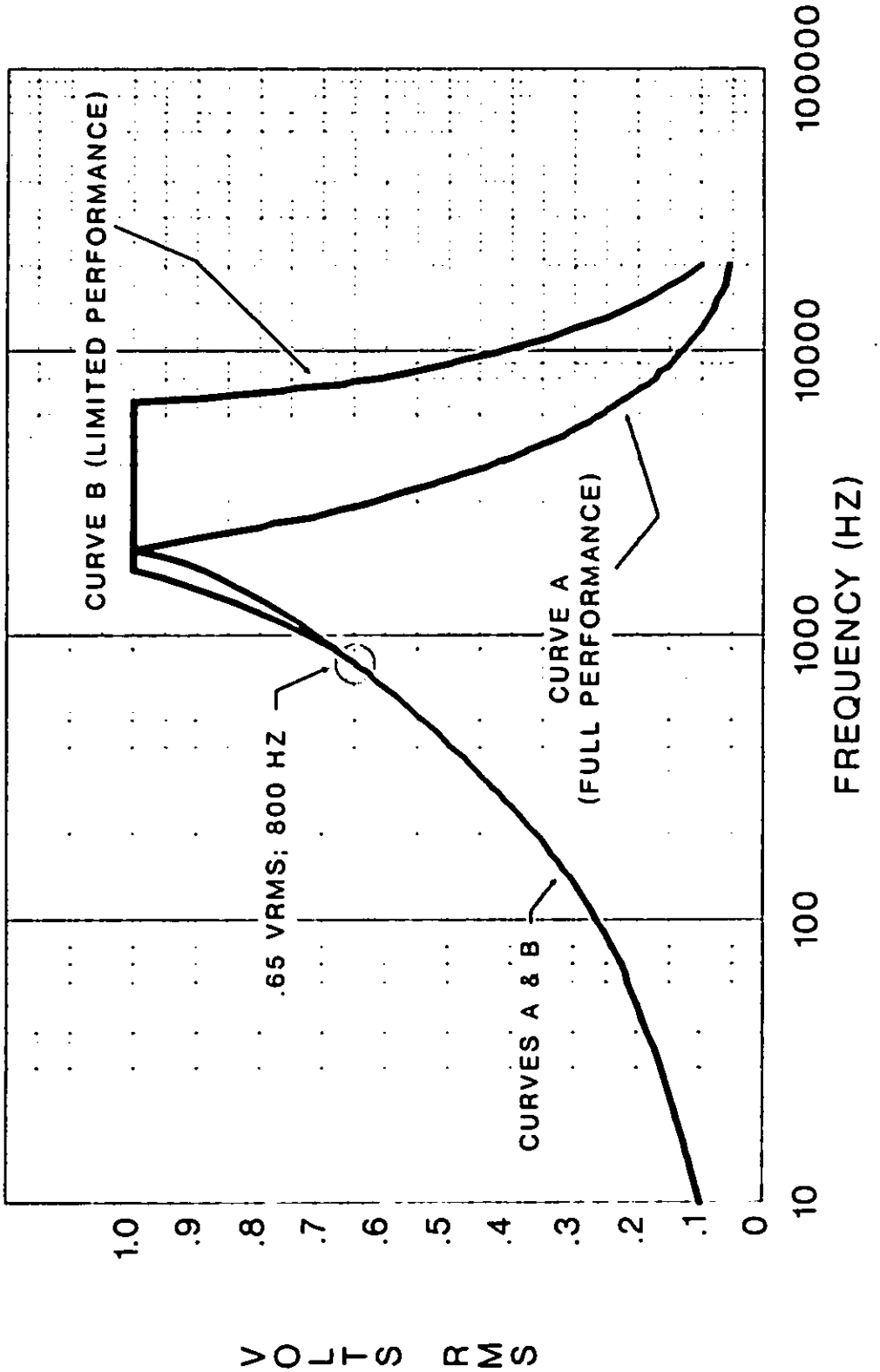


Figure 3. Output ripple frequency spectrum.

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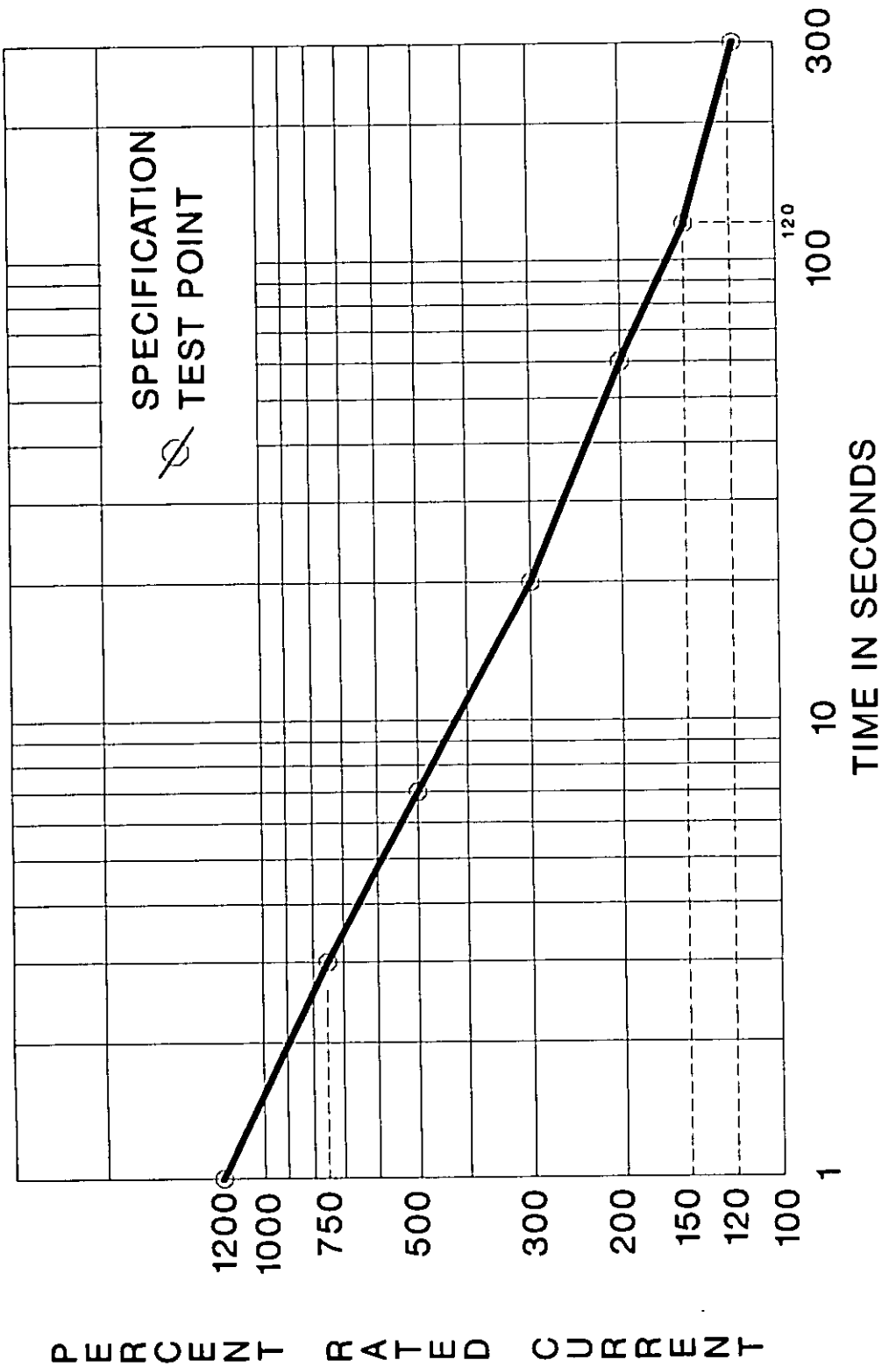


Figure 4. Converter current carrying capacity.

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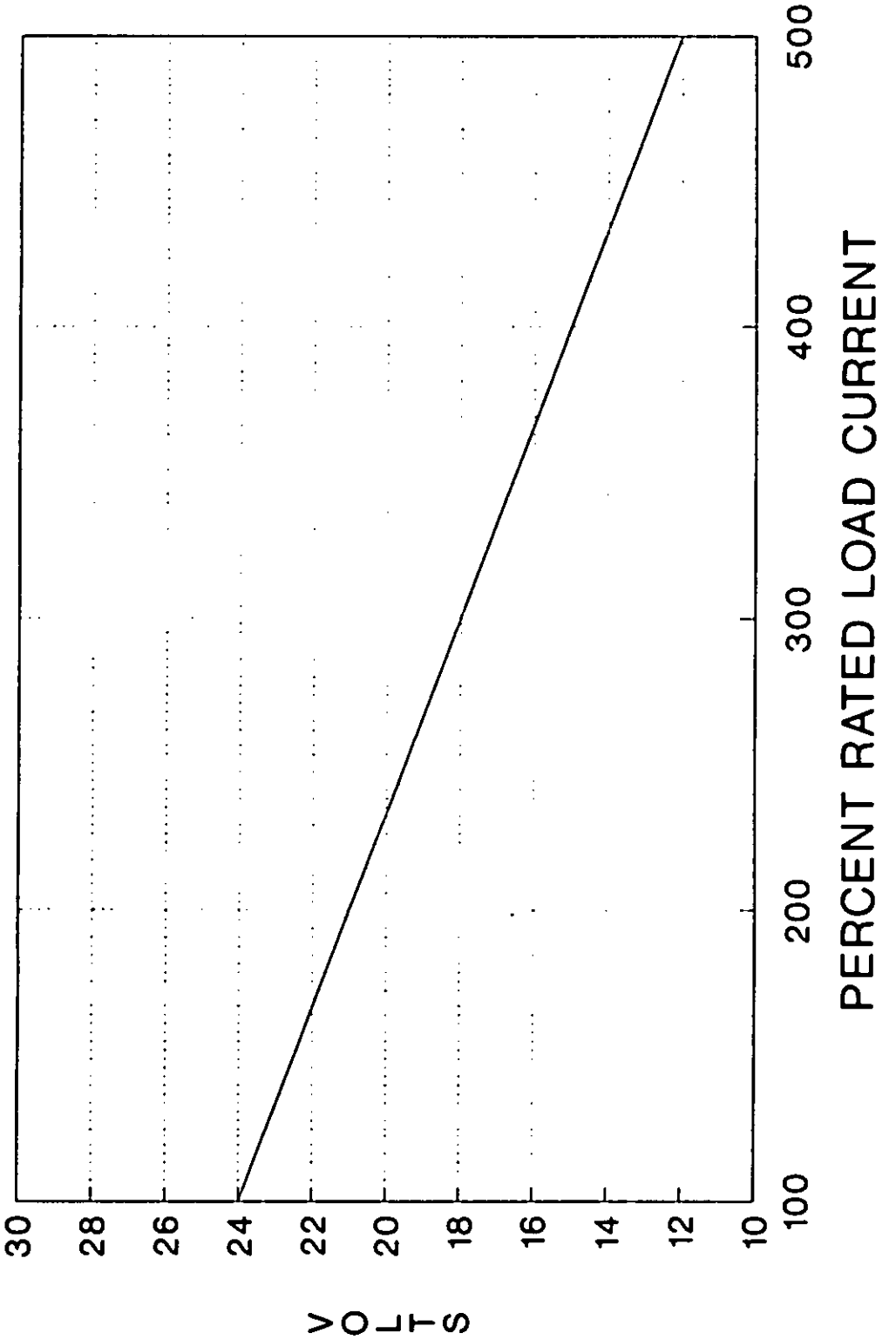


Figure 5. Minimum output voltage vs load.

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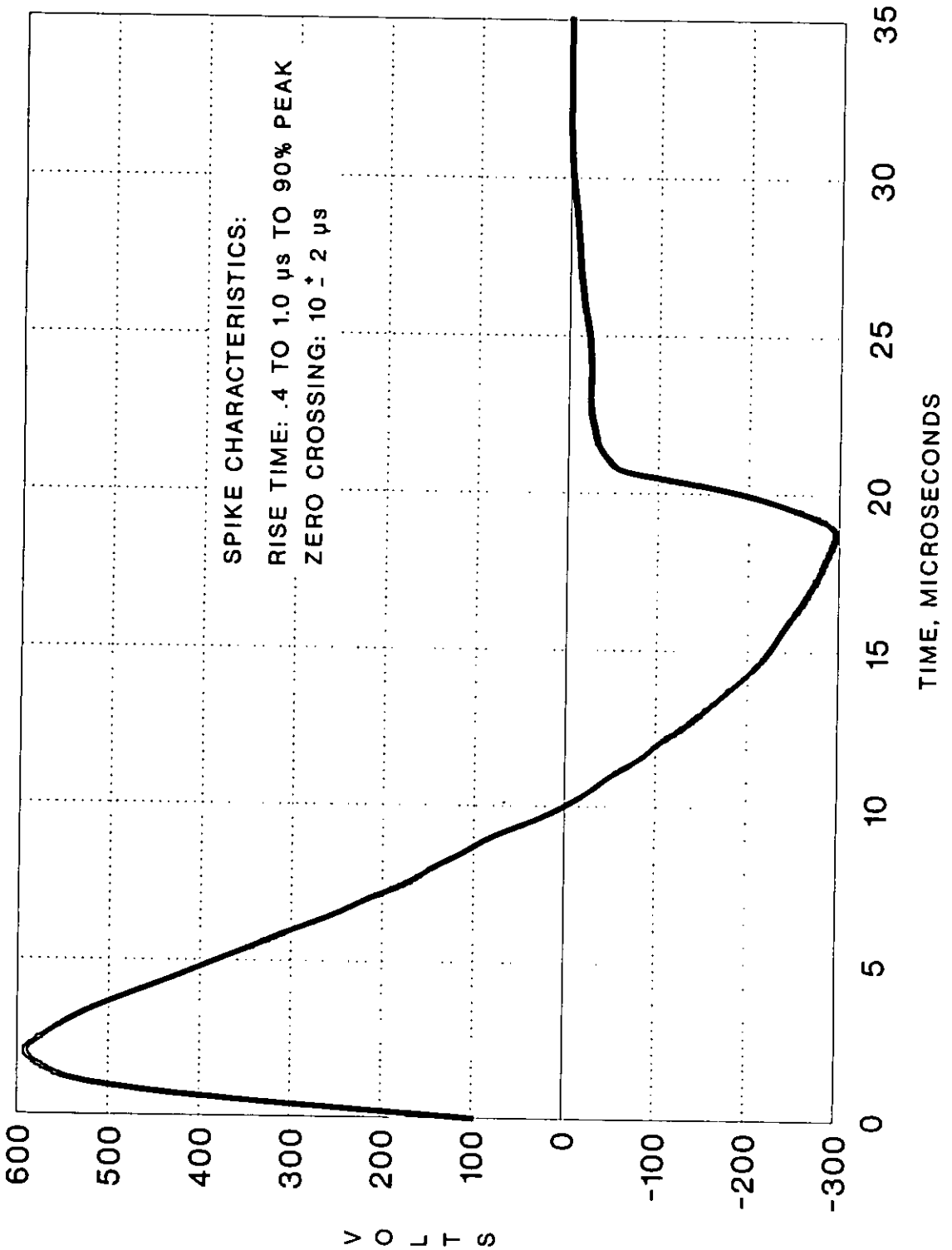
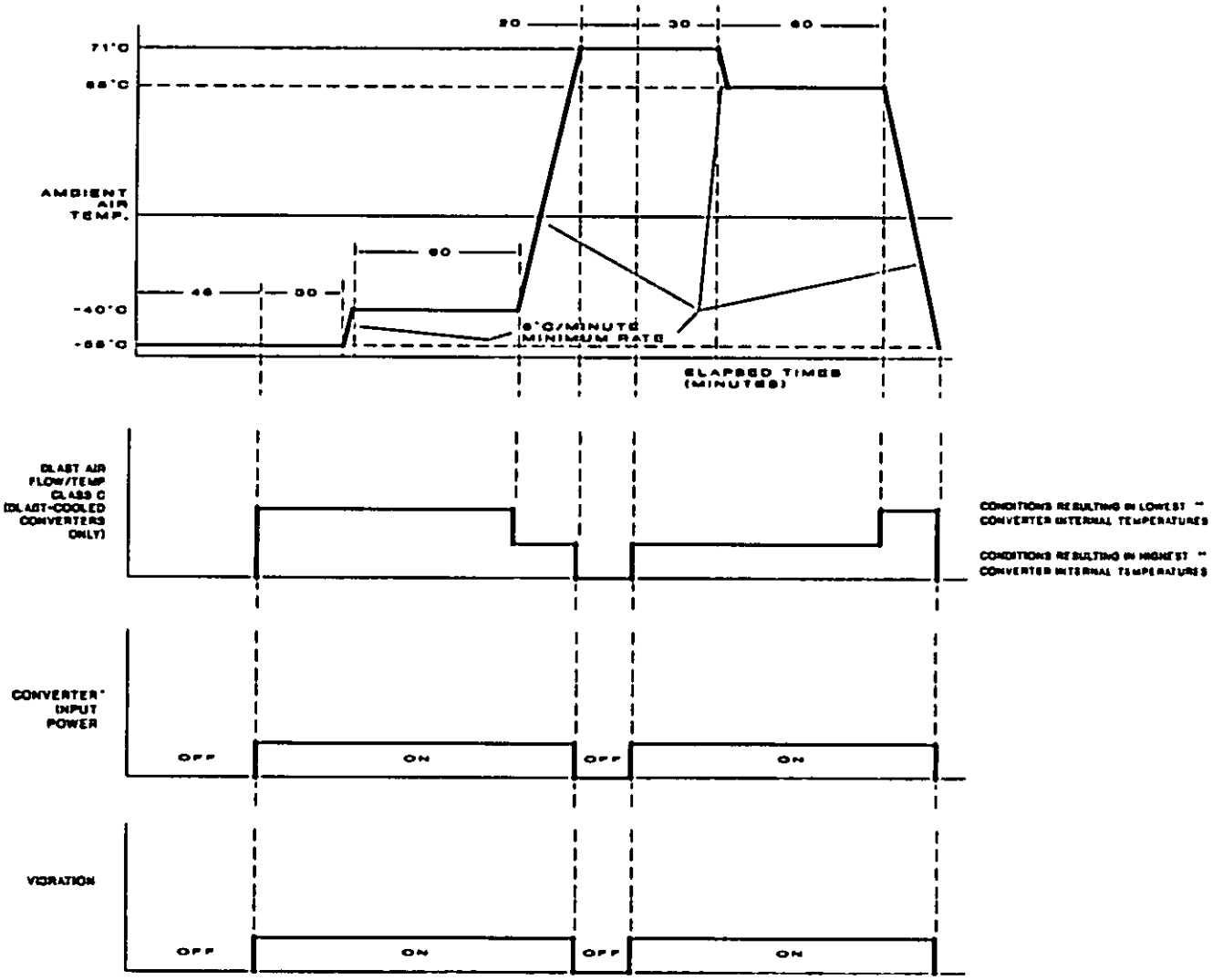


Figure 6. Spike susceptibility waveform.

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* ONE THIRD OF TOTAL CONVERTER OPERATING TIME SHALL BE CONDUCTED AT NOMINAL INPUT VOLTAGE, ONE THIRD AT MAXIMUM INPUT VOLTAGE, AND ONE THIRD AT MINIMUM INPUT VOLTAGE SPECIFIED IN THE APPLICABLE DETAIL SPECIFICATION.

** AS DETERMINED DURING THE TEST OF 4.7.2.2.1A.

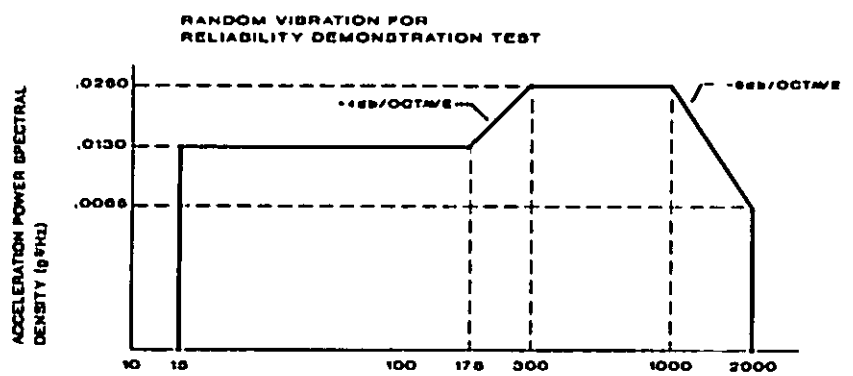


Figure 7. Reliability demonstration test profile.

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CONCLUDING MATERIAL

Custodians:
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Air Force - 99
DLA - GS

Review Activities:
Air Force - 80

Preparing Activity:
Navy - AS
Project No. 6130-0318

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-7115F	920918

3. DOCUMENT TITLE
Converters, Aircraft, Geneeral Specification for

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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Naval Air Systems Command (AIR-53632F)	(703) 692-2653	222-2653
c. ADDRESS (Include Zip Code) Power Systems Jefferson Plaza 2, Room 852 Washington, D.C. 20631-5360	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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